

DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 4 (AT1)

Chart Construction Test

Instructions:

For each question, write the program in the specified language (R or Python), generate the required

visualization, and answer the interpretation sub-questions clearly with justification.

CO2_AT1 – Chart Construction Test

Integrated Disciplines

Each question combines **Data Visualization + R Programming + another discipline** to promote interdisciplinary learning.

Question 1 (Healthcare Analytics)

Dataset

P1				Patient Age BMI Blood Pressure Disease Severity Low Medium High High Critical Critical Medium Low
P2				
P3	24	19.8	110	
P4				
P5	32	24.5	118	
P6				
P7	41	28.1	132	
P8				
	48	30.3	145	
	56	33.2	152	
	65	36.5	168	
	38	26.2	124	
	29	22.1	116	

Task

Construct a visualization in **R (ggplot2)** that:

- Maps **Age** to the x-axis.
- Maps **Blood Pressure** to the y-axis.
- Represents **Disease Severity** using an appropriate color palette.
- Represents **BMI** using point size.
- Justify why your aesthetic mappings and scales effectively communicate health risk. **Interdisciplinary Integration:** Healthcare + Medical Statistics + Data Visualization

Explanation:

Aesthetic Mapping

- X-axis → Age
 - Shows how blood pressure changes with age.
- Y-axis → Blood Pressure
 - Displays patient blood pressure values.
- Color → Disease Severity
 - Green → Low
 - Orange → Medium
 - Red → High
 - Dark Red → Critical
- Point Size → BMI
 - Larger points indicate higher BMI.

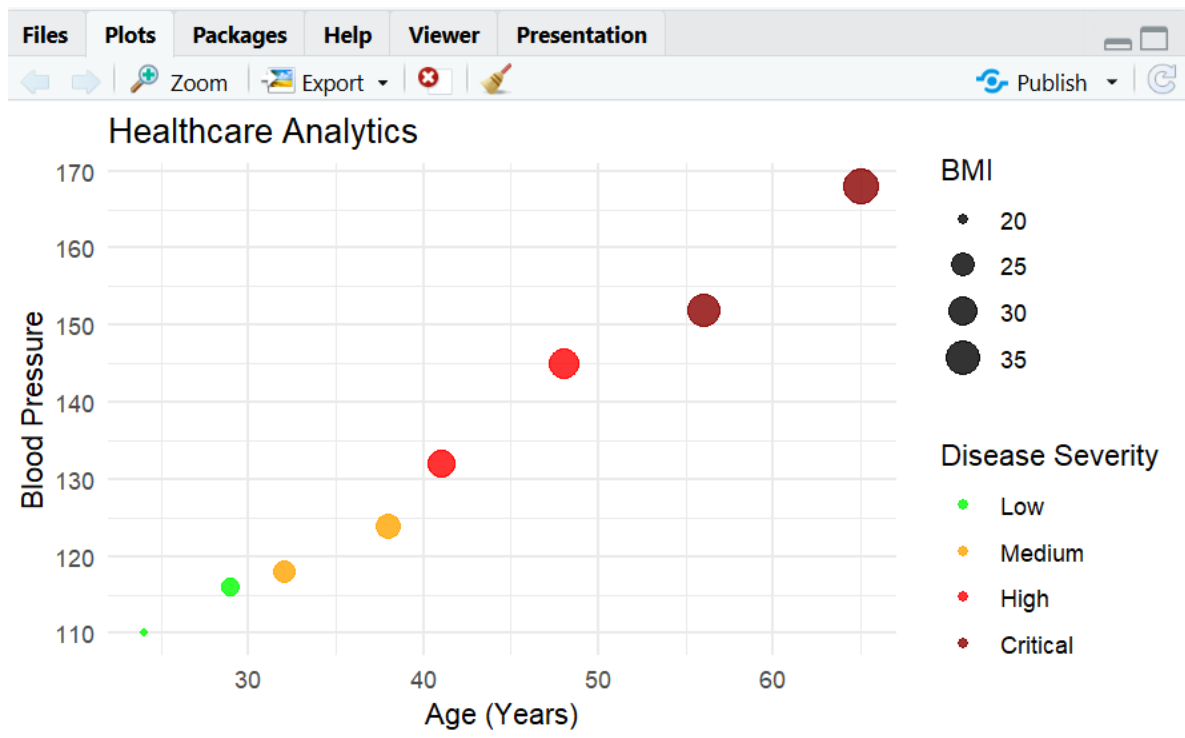
Code:

```
library(ggplot2)

health <- data.frame(
  Patient = c("P1", "P2", "P3", "P4", "P5", "P6", "P7", "P8"),
  Age = c(24, 32, 41, 48, 56, 65, 38, 29),
  BMI = c(19.8, 24.5, 28.1, 30.3, 33.2, 36.5, 26.2, 22.1),
  BloodPressure = c(110, 118, 132, 145, 152, 168, 124, 116),
  DiseaseSeverity = factor(c("Low", "Medium", "High", "High",
                             "Critical", "Critical", "Medium", "Low"),
                           levels=c("Low", "Medium", "High", "Critical")))
```

```
ggplot(health, aes(x=Age,
  y=BloodPressure,
  color=DiseaseSeverity,
  size=BMI)) +
  geom_point(alpha=0.8) +
  scale_color_manual(values=c("green","orange","red","darkred")) +
  labs(title="Healthcare Analytics",
  x="Age (Years)",
  y="Blood Pressure",
  color="Disease Severity",
  size="BMI") +
  theme_minimal()
```

Output:



Question 2 (Climate Science)

Dataset

Jan		Month Solar Radiation Temperature (°C) Rainfall (mm)	25
-----	--	--	----

Feb	180	20	18
Mar			12
Apr	200	23	8
May			5
Jun	240	29	65
Jul			150
Aug	280	35	170
Sep			120
Oct	320	39	85
Nov			45
Dec	300	36	30
	260	32	
	250	31	
	240	30	
	220	28	
	190	24	
	170	21	

Task

Develop **two visualizations**:

1. A Cartesian line chart showing temperature variation.
2. A Polar coordinate chart representing annual rainfall.

Compare the advantages and limitations of Cartesian and Polar coordinate systems for climate visualization.

Interdisciplinary Integration: Environmental Science + Mathematics + Data Visualization

Explanation:

Cartesian Chart

Advantages

- Easy to compare monthly temperature.
- Shows trends clearly.
- Suitable for time-series data.

Limitations

- Does not emphasize seasonal cycles.

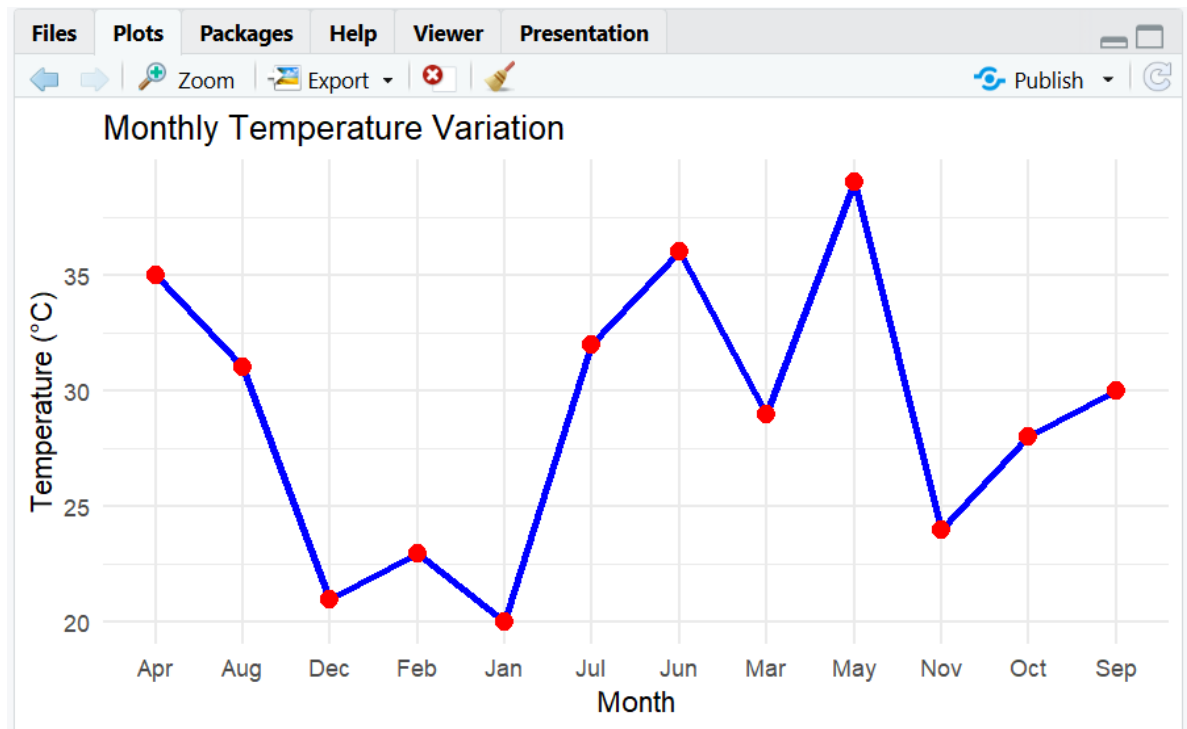
Code:

```
library(ggplot2)

climate <- data.frame(
  Month=c("Jan","Feb","Mar","Apr","May","Jun",
          "Jul","Aug","Sep","Oct","Nov","Dec"),
  SolarRadiation=c(180,200,240,280,320,300,260,250,240,220,190,170),
  Temperature=c(20,23,29,35,39,36,32,31,30,28,24,21),
  Rainfall=c(25,18,12,8,5,65,150,170,120,85,45,30)
)

ggplot(climate,
  aes(x=Month,
      y=Temperature,
      group=1)) +
  geom_line(color="blue", linewidth=1.2) +
  geom_point(color="red", size=3) +
  labs(title="Monthly Temperature Variation",
       x="Month",
       y="Temperature (°C)") +
  theme_minimal()
```

Output:



Polar Chart

Advantages

- Represents annual seasonal cycle.
- Circular layout shows repeating climate patterns.

Limitations

- Difficult to compare exact rainfall values.
- Less accurate for reading numerical differences.

Code:

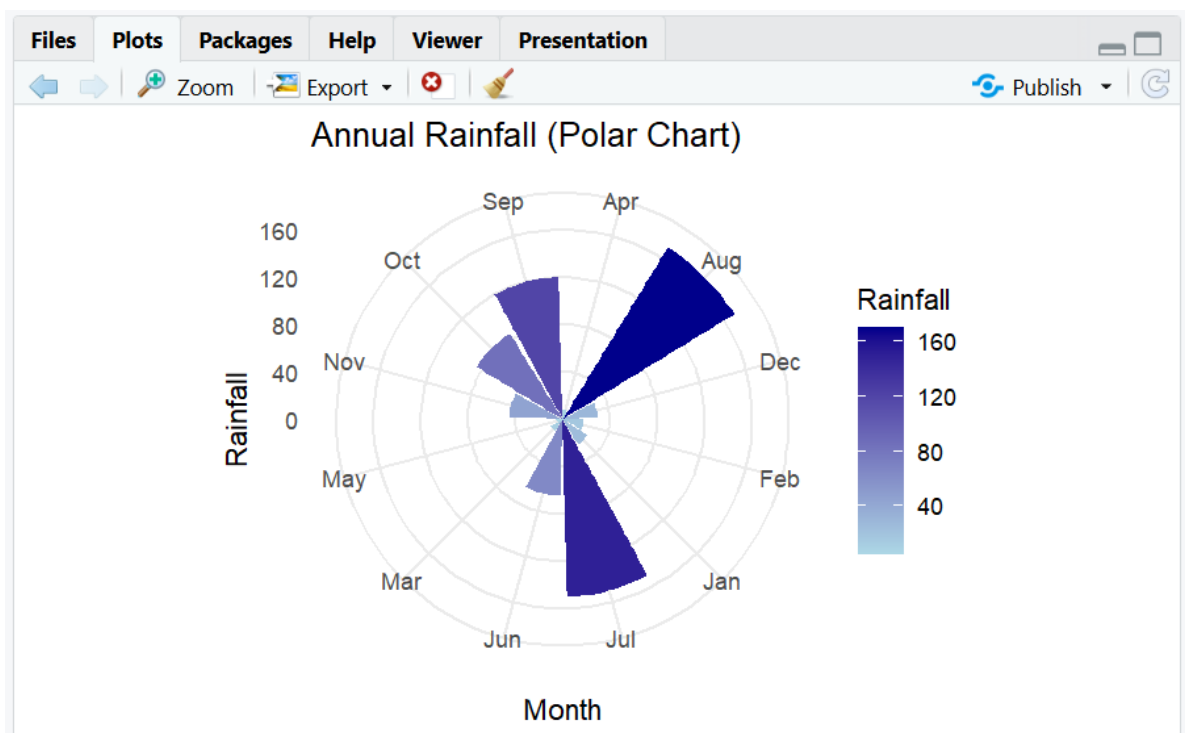
```
library(ggplot2)
climate <- data.frame(
  Month=c("Jan","Feb","Mar","Apr","May","Jun",
    "Jul","Aug","Sep","Oct","Nov","Dec"),
  SolarRadiation=c(180,200,240,280,320,300,260,250,240,220,190,170),
  Temperature=c(20,23,29,35,39,36,32,31,30,28,24,21),
  Rainfall=c(25,18,12,8,5,65,150,170,120,85,45,30)
)
ggplot(climate,
  aes(x=Month,
```

```

y=Rainfall,
fill=Rainfall)) +
geom_bar(stat="identity") +
coord_polar() +
scale_fill_gradient(low="lightblue",
                    high="darkblue") +
labs(title="Annual Rainfall (Polar Chart)") +
theme_minimal()

```

Output:



Question 3 (Business Analytics & Marketing)

Dataset

			Product Sales (₹ Lakhs)	Profit (%)	Market Share (%)	24
A						
B			20			
C	210	18	28			
D	185	14	15			

E	260	25	13
	145	12	
	120	10	

Task

Construct a visualization where:

- Sales are represented by bar height.
 - Profit percentage is represented using a sequential color scale.
 - Highlight the product with the highest market share using a contrasting color. Explain why your chosen color palette improves business decision-making.
- Interdisciplinary Integration:** Marketing + Business Intelligence + Visualization

Explanation

Mapping

- Bar Height → Sales
- Fill Color → Profit %
- Red Point → Highest Market Share

Color Palette

- Light color = Lower profit
- Dark green = Higher profit
- Red marker quickly identifies market leader.

Benefits

- Compare sales easily.
- Understand profitability.

Code:

```
library(ggplot2)

business <- data.frame(
  Product=c("A","B","C","D","E"),
  Sales=c(210,185,260,145,120),
  Profit=c(18,14,25,12,10),
```

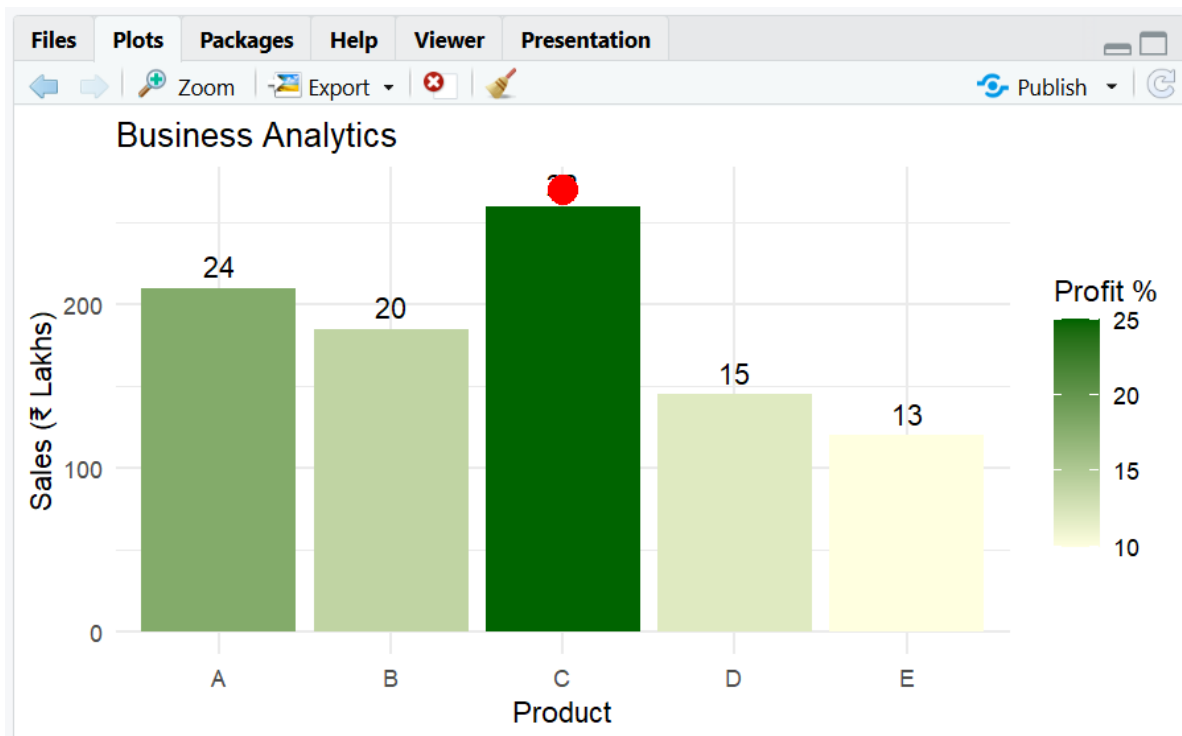


```

MarketShare=c(24,20,28,15,13)
)
business$Highlight <- ifelse(business$MarketShare==
                             max(business$MarketShare),
                             "Highest","Others")
ggplot(business,
       aes(x=Product,
           y=Sales,
           fill=Profit)) +
  geom_bar(stat="identity") +
  geom_text(aes(label=MarketShare),
            vjust=-0.5) +
  geom_point(data=subset(business,
                         Highlight=="Highest"),
            aes(y=Sales+10),
            color="red",
            size=5) +
  scale_fill_gradient(low="lightyellow",
                     high="darkgreen") +
  labs(title="Business Analytics",
       y="Sales (₹ Lakhs)",
       fill="Profit %") +
  theme_minimal()

```

Output:



Question 4 (Transportation Engineering)

Dataset

Direction	Vehicles/hour 980
North	1240
North-East 760	1420
East	860
South-East 890	
South	
South-West 1130	
West	

Direction North-West 720	Vehicles/hour

Task

Using **R Programming**:

- Construct a **Polar Coordinate Chart** representing traffic flow.
- Apply an appropriate color palette to distinguish traffic density.
- Explain why a Polar chart is more suitable than a Cartesian bar chart for directional traffic analysis.

Interdisciplinary Integration: Civil Engineering + GIS + Transportation Analytics

Explanation

Mapping

- Direction → Angular position
- Vehicles/hour → Bar length
- Color → Traffic density

Why Polar Chart?

Advantages

- Naturally represents directions.
- Similar to a compass.
- Easy to identify busiest directions.
- Better than Cartesian charts for directional data.

Cartesian Limitation

- Directions appear as simple categories.
- Does not represent compass orientation.

Code:

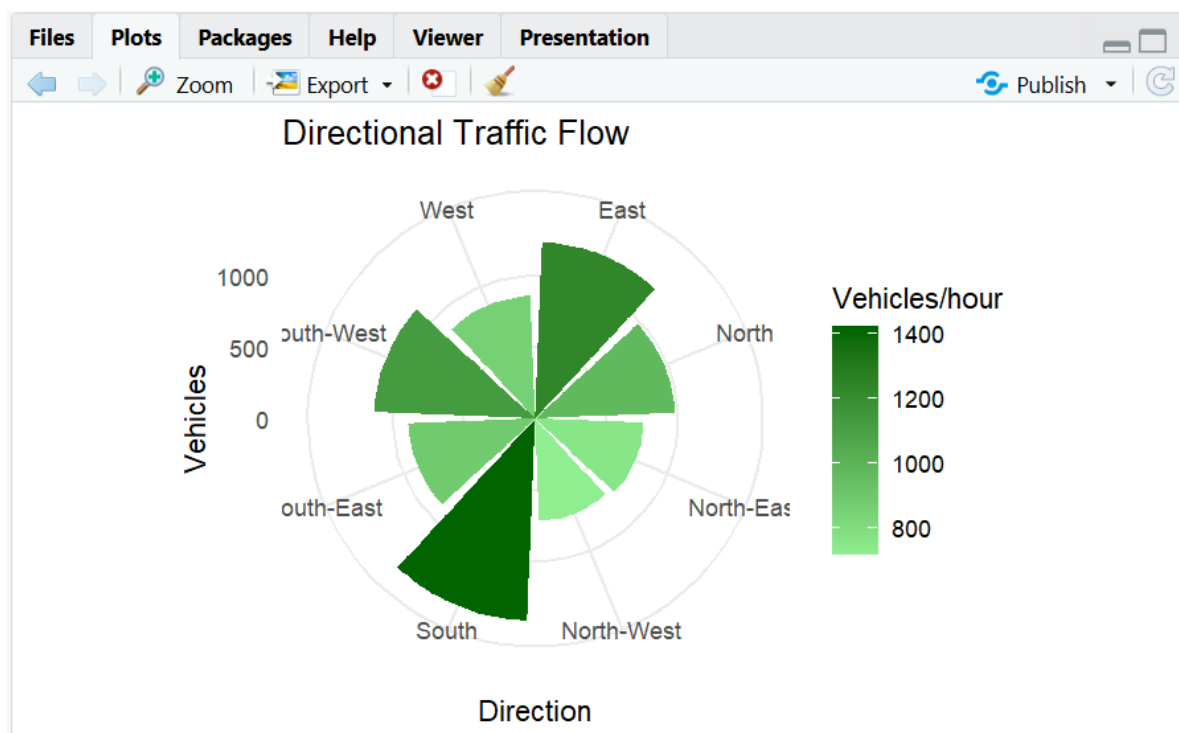
```
library(ggplot2)
traffic <- data.frame(
  Direction=c("North","North-East","East",
             "South-East","South",
             "South-West","West","North-West"),
```

```

Vehicles=c(980,760,1240,890,1420,1130,860,720)
)
ggplot(traffic,
  aes(x=Direction,
      y=Vehicles,
      fill=Vehicles)) +
geom_bar(stat="identity") +
coord_polar() +
scale_fill_gradient(low="lightgreen",
                    high="darkgreen") +
labs(title="Directional Traffic Flow",
      fill="Vehicles/hour") +
theme_minimal()

```

Output:



Question 5 (Environmental Sustainability Dashboard)

Dataset

A				City AQI Green Cover (%) CO ₂ Emission Population (Lakhs) 9
B				
C	48	45	1.8	
D				
E	82	32	3.5	
F				
	135	21	5.6	
	165	18	7.4	
	72	38	2.9	
	95	30	4.2	

Task

Design a visualization dashboard that includes:

- Appropriate aesthetic mappings.
- Suitable scales for continuous variables.
- An effective color palette for AQI categories.
- Justification for every visualization design choice.

Explanation

Visualization 1

- Bar chart compares AQI across cities.
- Color categories:
 - Green → Good
 - Yellow → Moderate
 - Orange → Poor
 - Red → Very Poor

Visualization 2

- X-axis → Green Cover (%)
- Y-axis → CO₂ Emission

- Point Size → Population
- Color → AQI Category

Suitable Scales

- Continuous scale for:
 - Green Cover
 - CO₂ Emission
 - Population
- Discrete color scale for AQI categories.

Code:

```
library(ggplot2)

city <- data.frame(
  City = c("A","B","C","D","E","F"),
  AQI = c(48,82,135,165,72,95),
  GreenCover = c(45,32,21,18,38,30),
  CO2 = c(1.8,3.5,5.6,7.4,2.9,4.2),
  Population = c(9,15,22,28,12,18))

city$AQI_Category <- cut(
  city$AQI,
  breaks = c(0,50,100,150,200),
  labels = c("Good","Moderate","Poor","Very Poor"))

ggplot(city,
  aes(x = City,
      y = AQI,
      fill = AQI_Category)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("green","yellow","orange","red")) +
  labs(title = "AQI")

ggplot(city,
  aes(x = GreenCover,
      y = CO2,
```

```

size = Population,
color = AQI_Category)) +
geom_point(alpha = 0.8) +
scale_color_manual(values = c("green", "yellow", "orange", "red")) +
labs(title = "Green Cover vs CO2")

```

Output:

